

CHAPTER 3

Characteristics, advances, and challenges in blockchain-enabled cyber-physical systems

Manohar Sai Burra and Soumyadev Maity

Department of Information Technology, Indian Institute of Information Technology Allahabad,
Prayagraj, Uttar Pradesh, India

1. Introduction

With advancements in science and technology, the world is becoming a global village. At present, there is tremendous growth in the digital transformation of many products and activities. The days have transformed from people connecting through the internet toward any physical object with software embedded connecting to the internet. We now call this paradigm the internet of things (IoT). The integration of physical processes, computation, the internet, and control of these physical objects through actuators is called the cyber-physical system (CPS) (Sztipanovits et al., 2011). CPS is a feedback system where the data from physical objects affect computation, and this results in affecting the control and vice versa (Derler et al., 2013). IoT, CPS, and the industrial internet of things (IIoT) have many overlapping features and characteristics in design, deployment, and management.

CPSs play a crucial role in improving the performance of many products and activities in areas such as next-generation mobile systems, vehicular networks (VANET), smart homes, and industrial automation. With the increased use of CPS technology, it is important to secure communication and control of the system (Cardenas et al., 2009). CPSs are usually deployed with the internet as the backbone network. So the CPS should withstand cyberattacks coming through the internet. The CPS architecture focuses on building resilient systems that recover and provide services at any instance of a cyberattack (Zheng et al., 2017).

CPSs are a complex integration of physical processes, computation, and actuators of resource-constrained devices, which results in newer challenges in designing, deploying, and managing these systems. The prime focus while realizing a CPS should be on measuring the system's resilient capabilities, reviewing the quantitative and qualitative factors for assessing the system, and providing cyber resilience strategies per the assessment (Ding et al., 2015).

The CPS involves complex interactions of various processes between the cyber and physical world. CPS involves multiple disciplinary approaches that merge computer science, mechatronics, electronics, cybernetics, mathematics, logic, and more. These interactions bring in newer complexities to the CPS system design and development. To achieve reliability for such systems is a tough task. Blockchain technology has become a boon to establish consensus and achieve reliability among these processes of the CPS (Bach et al., 2018). Most of the blockchain-related consensus algorithms employ proof-of-work (PoW) or proof-of-stake (PoS) or a hybrid of these techniques. PoW requires a prover party to perform a certain moderately hard but feasible task (computation) that must be easy for the verifier to check the proof or vice versa. PoS requires that an individual can validate or mine block transactions based on available assets such as coins in coin mining.

Blockchain has gained popularity as a secure electronic distributed ledger since the introduction of Bitcoin. But it has progressed far beyond digital currency and into other areas of application. Many domain areas requiring the storage and retrieval of transaction data use blockchain technology. One of the important types of blockchain technology is smart contracts, which allows essential automation of workflow in many systems. This smart contract allows two entities to execute an agreement that will be irreversible, transparent, and traceable.

The main contributions of this chapter are the following:

1. We have reviewed and provided a detailed comparison of various existing blockchain-enabled CPS schemes based on different characteristic properties.
2. We have identified various challenges in blockchain-enabled CPSs.
3. We have reviewed and discussed the possible security attacks and the likelihood of blockchain-enabled CPSs.
4. Additionally, we have also provided a brief introduction to blockchain and CPSs.

The rest of the paper is organized as follows: Sections 2 and 3 discuss the background for both CPS and blockchain in detail. Section 4 describes various blockchain-enabled CPS frameworks. Section 5 provides a comparative analysis of various existing blockchain-enabled CPS schemes based on different characteristic properties. Section 6 states the various technical and nontechnical challenges and some specific security attacks. Finally, we conclude the chapter in Section 7.

2. Background of CPS

With the increase in the number of digital devices globally, there is a growing demand for CPS-based applications worldwide. It requires accurate coordination and a combination of both computational and physical resources to deliver performance. It plays a crucial role in transforming critical infrastructure systems such as traffic management, personalized healthcare, autonomous automobiles, smart grids, the industrial internet, the IoT, and many others through innovative, intelligent services. In CPS, there is complex information transfer and exchange among various physical processes. These complex interactions require consensus among the processes for the reliable and efficient performance of critical infrastructure systems. This section discusses various use cases in CPSs integrated into the networks. We discuss the following scenarios: (1) healthcare systems, (2) transportation and vehicular ad-hoc networks, (3) urban infrastructure, and (4) defense systems.

2.1 Healthcare systems

In the healthcare sector, CPS enables the development of newer medical devices that are implantable equipment and wearable sensors that collect health data from humans to form wireless body area networks. An effective health data analysis can provide better timely disease diagnosis and improve the overall quality of life. Further, these personal healthcare devices would assist hospitals in providing economically viable services (Haque et al., 2014; Kurde et al., 2019; Zhang et al., 2015).

Fig. 3.1 shows a CPS-enabled healthcare system where the user wears a sensor device that collects health data and transfers it to the connected mobile device. The mobile device forwards the data securely to a cloud server through the internet. The various other entities such as the doctor can obtain health data to perform diagnoses or the emergency services can be called automatically when continuous data analysis shows signs of an emergency health issue.

2.2 Transportation and vehicular ad-hoc networks

In the automobile industry, there is advancement in the integration of communication and computation technologies into vehicles. Autonomous and smart vehicles can be built that will reduce the load on human driving and bring increased safety and ease of travel. Further, CPS can boost growth in smart highways, vehicular networks, railroad systems, and airspace management in the aviation sector (Abid et al., 2011; Mehmood et al., 2020).

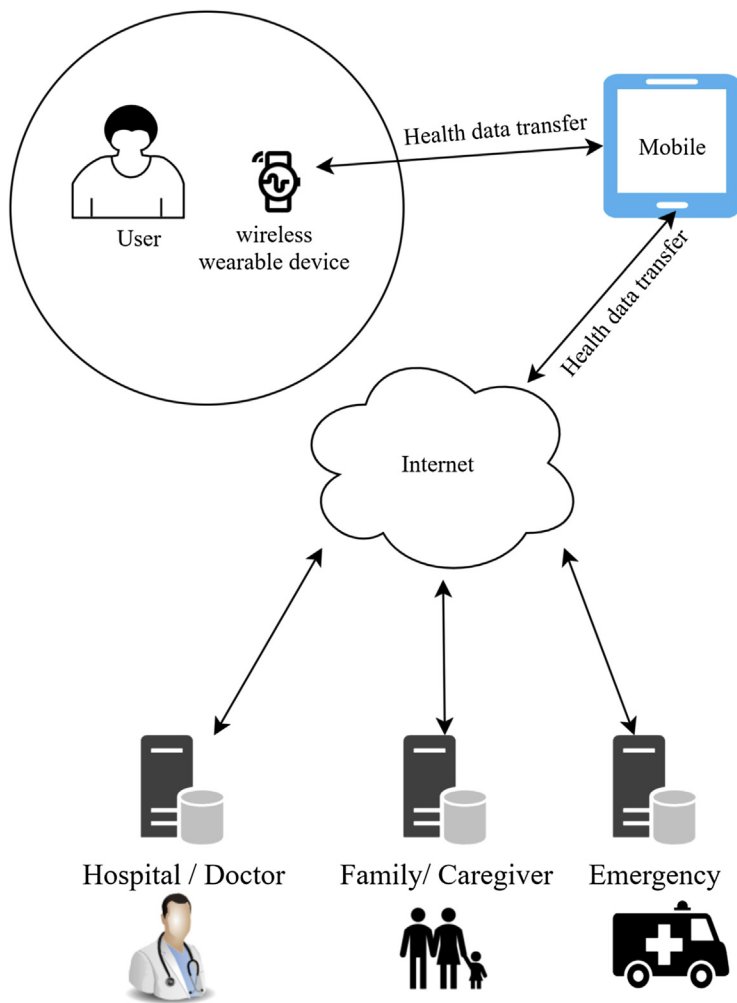


Figure 3.1 CPS-enabled healthcare system.

Fig. 3.2 shows a vehicular ad-hoc network. There are three types of communication: (1) vehicle to vehicle (V2V) communication, (2) vehicle to infrastructure (V2I) communication, (3) infrastructure to infrastructure (I2I) communication. CPS technology provides benefits to improve traffic congestion through embedding sensors onto vehicles that continuously collect data and upload onto the server for traffic analysis, and these servers provide control data for interactive traffic control systems, ultimately preventing accidents and reducing traffic congestion.

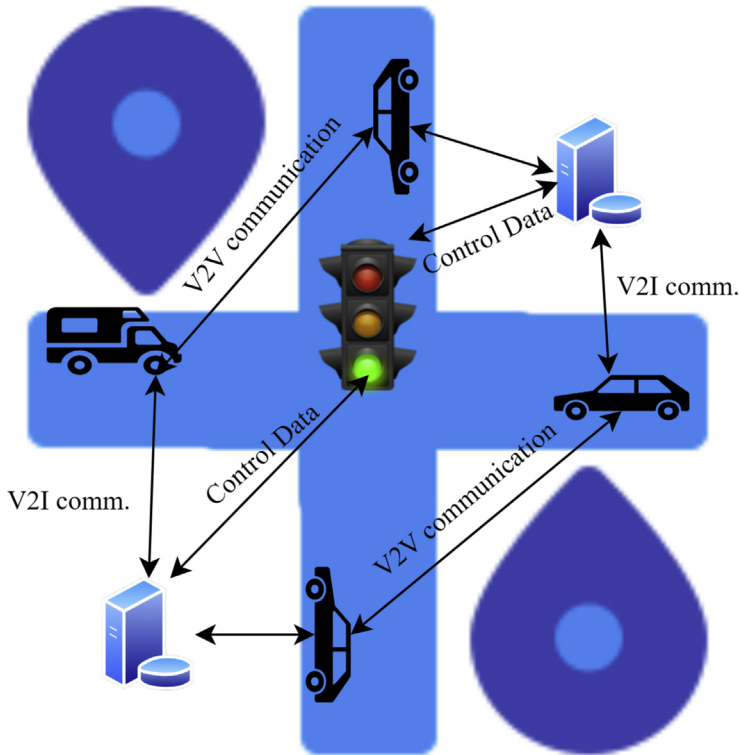


Figure 3.2 CPS-enabled transportation and VANET.

2.3 Urban infrastructure

In civil infrastructure development such as dams, sewage management, wastewater treatment, and many more examples, CPS can provide continuous monitoring, assessment, and control over these systems. Smart grids can be built to treat wastewater and manage water resources efficiently. In a large bridge or dam, CPS can provide early warning signs on the structural damages of infrastructure and thus save lives and prevent property damage (Saetta et al., 2019; Sun et al., 2018).

Fig. 3.3 shows a water treatment and management system where the water from natural water body sources is treated to provide drinking water to users. The computing server and control system will monitor, analyze, and send control data to other entities to improve the performance and efficiency of the water treatment system.

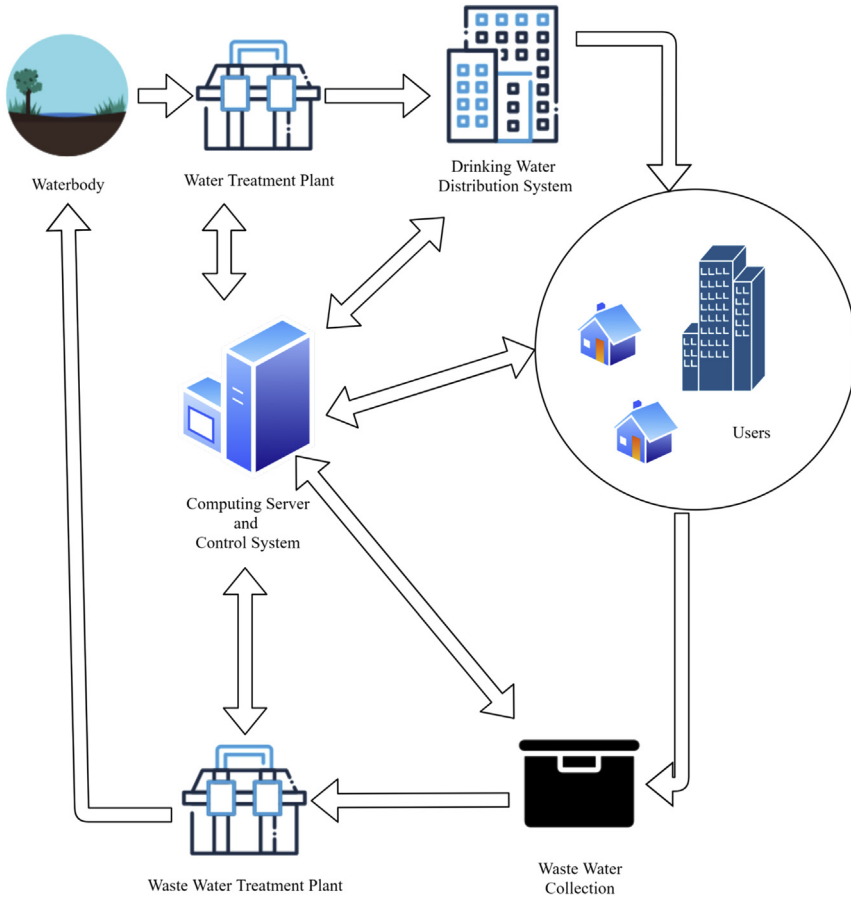


Figure 3.3 CPS-enabled water treatment and management system.

2.4 Defense systems

Warfare can greatly damage both the living and their property. Today nations are striving to create tools and technology to win wars with less damage. CPS can enable in development of smart precision-guided weapons, smart unmanned vehicles, and wireless wearable computing devices. This will reduce the number of soldiers involved in direct combat but increase remote warfare capabilities (Gill, 2008).

Today, individuals and businesses are directly or indirectly utilizing these systems. There is an increasing demand for control and automation in complex processes, specifically where human life is in danger while performing hazardous tasks. We could employ CPSs to eliminate the human need to

perform such hazardous tasks as investigating volcanic sites, detecting mining areas, or handling radioactive elements. Further, industries utilizing a CPS could improve their overall efficiency, performance, and economy.

3. Background of blockchain

Blockchain is the underlying concept behind Bitcoin cryptocurrency (Nakamoto and Bitcoin, 2008). Today, there are hundreds of cryptocurrencies that employ blockchain to create, store, and verify transactions (Corbet et al., 2018). Blockchain is a distributed and decentralized database, so it differs from the database of a traditional financial instrument (Sarmah, 2018). In the early stages, the role of blockchain was limited to its application in creating cryptocurrencies. Today, blockchain finds its use in many newer areas, including and not limited to healthcare, smart contracts, the IoT, CPS, and IIoT (Alladi et al., 2019; Dedeoglu et al., 2020; Panarello et al., 2018).

Fig. 3.4 shows the blockchain structure of Bitcoin. Each block is a composition of the previous block hash, timestamp, nonce, and Merkle root of all the transactions in that block. These blocks are interlinked together to form a blockchain. The first block of the blockchain is called the genesis block, and it does not contain any previous hash value. The blockchain is a secure ledger of verifiable records. A new block can only be

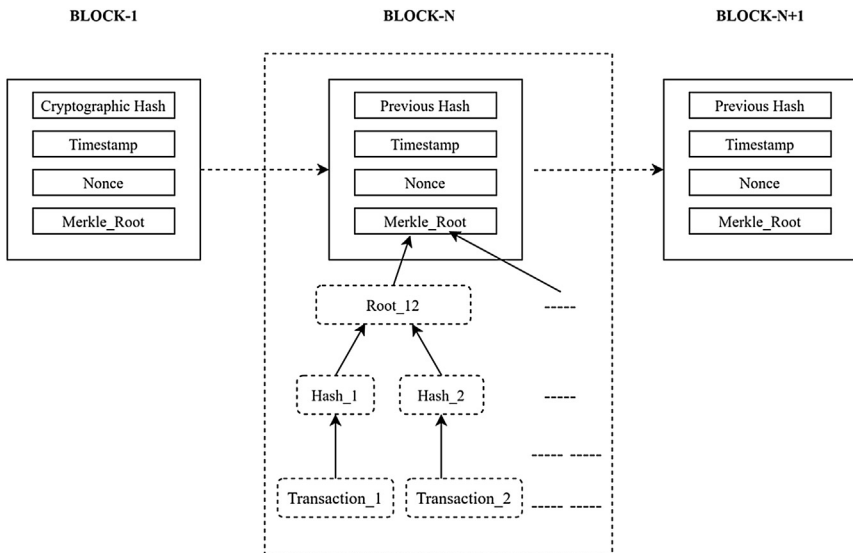


Figure 3.4 Blockchain structure.

added through a PoW mechanism and this requires solving for a pattern in the hash and thus may require high computational power. The PoW itself takes significant time compared to the verification of PoW, which takes very little time. Any malicious entity that wants to replace a block in the chain must solve for subsequent resulting chains in the blockchain, and this could take exponential time so cannot realistically be performed. Therefore the blockchain is a tamperproof ledger of records.

3.1 Types of consensus algorithms

Blockchain technology is one of the important techniques to build consensus algorithms. These algorithms are responsible for achieving reliability even in a complex system with high interactions and faulty processes. Two of the methods to achieve consensus are discussed next (Bach et al., 2018; Mingxiao et al., 2017):

- **Proof-of-work:** As described earlier, PoW involves entities called miners to solve a puzzle. The solution to the puzzle is called the nonce. Here, the miners can take advantage of distributed parallel computation to achieve the solution faster. Once the solution is available, the miners publish the nonce value to claim the rewards. Generally, in blockchain-based cryptocurrency, the currency is given as the reward.
- **Proof-of-stake:** The PoS mechanism was created as an alternative mechanism to reduce the computational power required to add a block to the blockchain. The PoS allows miners to mine with respect to the stake they hold. For example, a miner with a 3% stake can only mine 3% of the blocks in the blockchain. This allows miners with good stakes to mine more and those with less to mine fewer. This promotes miners to increase stakes to mine the blocks.

3.2 Essential features of blockchain technology

- **Immutability:** Many of the blockchain users maintain full nodes. This is quite different from a traditional centralized ledger. Any addition or validation of the block requires consensus from the majority of the users. Moreover, a block in the chain cannot be replaced since it requires changing the other blocks in the chain. This process requires exponential time, so the ledger is immutable.
- **Decentralization:** The immutability and decentralization go hand in hand. The blockchain ledger is not owned by any central authority. Many users have a copy of all the blocks of the blockchain, and any

entity maintaining a complete ledger is called a full node. As no single entity governs the blockchain, the fear of monopoly is eliminated, creating trust among all the participating entities of the blockchain. Most of the work involved in building a blockchain can be automated, thereby reducing the risk of human error. The blockchain technology employed in cryptocurrency and asset management can provide users with complete control over their coins and assets without the need for any third-party entity involvement. Through decentralization comes transparency, since any transaction in the blockchain can be validated with only public values.

- **Improved security:** Blockchain technology employs cryptographic techniques to achieve strong security. It employs encryption and signature mechanisms to accurately validate transactions in the blockchain. The core technique used in the blockchain is hashing, where the hash function employed is a secure one-way hash function that transforms original data to a different form to protect the actual data as well as enabling easier verification and processing.
- **Privacy:** Blockchain technology inherently provides privacy for user information. The ledger, which is available to all the entities, does not contain sensitive information of its users. It usually contains public data, which does not create privacy concerns for the users. Any user can perform a transaction without the need to provide personal, sensitive information.

4. Blockchain-enabled cyber-physical systems

In this section, we will discuss the advancement of blockchain technology in various CPS-enhanced areas.

4.1 Supply chain management

The adoption of blockchain in supply chain management (SCM) is growing rapidly. These SCM systems have a large number of participating entities, which creates complex interactions, so requires better mechanisms to address the various issues involved. The integration of blockchain in the SCM systems provides mechanisms to reduce operational costs, improve quality, and create sustainable and flexible systems. The primary use of blockchain in these SCM systems traces back to accountability and transparency. With CPS enhancing SCM, various components such as sensors, integrated circuits, global positioning tags, barcodes, and radio-frequency identification tags can be used to

track the location of the shipments, parcels, and products. This brings out techniques to efficiently track real-time goods from source to destination (Blossey et al., 2019; Saberi et al., 2019).

Queiroz and Fosso Wamba (2019) have shed light on the behavior of various entities involved in SCM across India and the United States on the adoption of blockchain. They have proposed a novel model that established a clear difference of behaviors of adopters in both India and the United States. Montecchi et al. (2019) developed a framework for SCM to address the issues of the provenance of goods through the integration of blockchain. Their framework contained a wheel of four assurances: the first is the origin of goods assurance; the second is custody of assurance, which enables a consumer to track the goods throughout the entire supply chain; the third is integrity assurance; and finally, the fourth is authenticity assurance. In Toyoda et al. (2017) the authors have proposed blockchain-based techniques to fight counterfeiting in post-SCM.

Fig. 3.5 shows the SCM network of apple juice beverage production. Here, blockchain can be employed at each step of the process, where the ledger data from every step can be later verified to identify the location of any entity involved. For example, if a batch of apple juice produced has been tampered with, then we can use the help of the ledger to track back into every step of the production process to identify the possible location of tampering. This will help the SCM industry to trace and audit the systems.

4.2 Healthcare sector

Due to the ever-growing population, there is an ever-growing demand for healthcare services across the world. The healthcare sector requires stringent mechanisms to protect patient data confidentiality. In the world of interconnected smart devices and people, there is a greater need to secure communication and provide privacy of data at every instance of health data management. There are some distinctive requirements for the healthcare sector, and these include but are not limited to secure health data transfer, authentication of patient data, and interoperability. Many healthcare institutions worldwide maintain electronic health records of patients. These records are either at the hospital repository or stored at a third-party server. There exist mechanisms for multiple institutions to access patient data on a shared basis, but there are very few mechanisms to protect privacy. There is a need for patient-centric health data sharing that includes security and privacy (Hathaliya et al., 2019; Kassab et al., 2019; McGhin et al., 2019; Zhang et al., 2017).

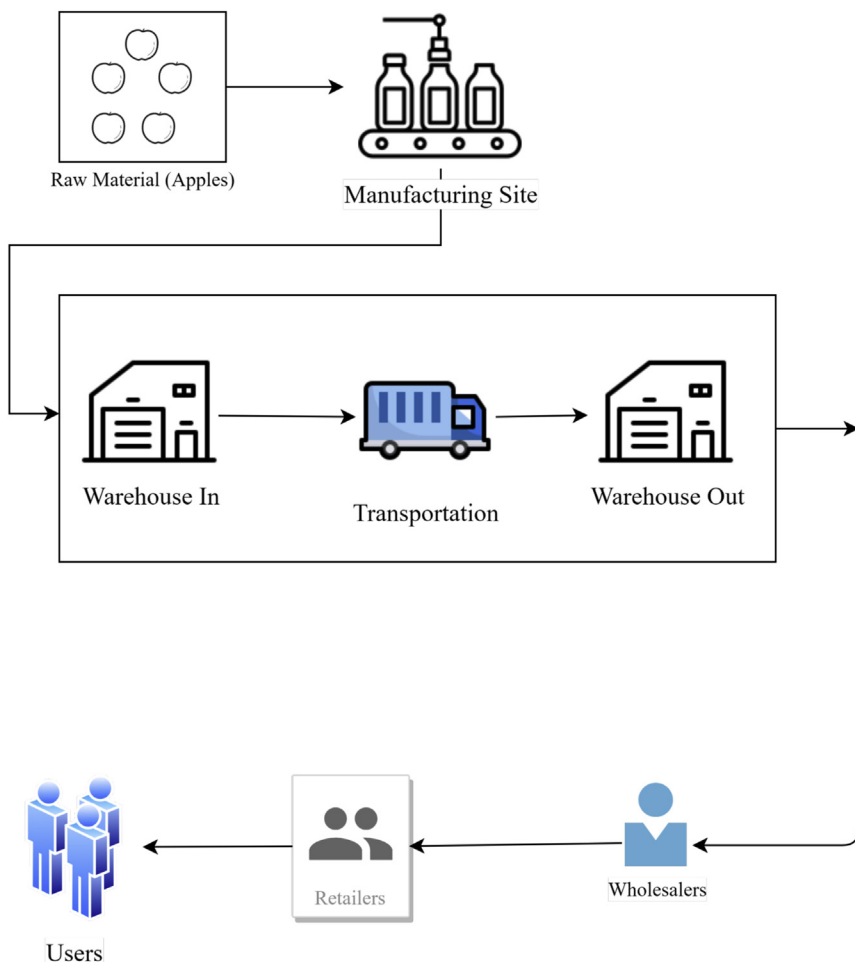


Figure 3.5 Supply chain management of apple juice beverage.

Blockchain can provide decentralized storage of medical records, which enhances the quality of data and enables faster access time and patient-driven interoperability. [Li et al. \(2018\)](#) proposed a medical data preservation system using blockchain technology, where if any record is found to be tampered with, it can be easily verified. They even have employed cryptographic techniques to secure sensitive user information. [Yue et al. \(2016\)](#) proposed a patient-driven health data monitoring and control that protects the privacy of users. The authors have implemented the same through a user interface application. They also provide mechanisms to employ third-party entities for computation purposes without any privacy issues through

the use of multiparty computation. [Ekblaw et al. \(2016\)](#) have proposed a prototype for medical health data handling and use. They provided mechanisms to allow hospitals and patients to give metadata from the health records for big data analysis without impacting data privacy. The data analysis can provide insights into the health status of various patients and diseases.

In the insurance industry, blockchain can enable a ledger of immutable records, where the blocks contain data related to a set of health-related events as they took place, and this allows easier verification of events. Further, no entity can change the sequence of events in the blockchain nor change the data in the blockchain, thus giving both the insurance company and the insured better mechanisms of the claim process. [Raikwar et al. \(2018\)](#) have identified various blockchain-enabled use cases for insurance processes. They have utilized smart contract mechanisms to simulate real-world insurance policies across multiple endusers. [Gatteschi et al. \(2018\)](#) elaborated on the possibilities of blockchain integration in the insurance sector considering various factors that are involved.

4.3 Emerging industrial control systems

Today, we have a new emerging market in an industrial sector called the industrial internet of things where the physical components of the industry are enabled with intelligent software systems that can be controlled through the internet. Moreover, there is a great demand to automate the processes through the use of computing capabilities. Today, the IIoT has grown larger than imagined a decade ago, where every piece of equipment in the industrial sector can be integrated into the IIoT. These can range from machines in manufacturing industries to sensors nodes in wireless networks. The IIoT network architecture is not standardized, distributed, nor open in nature. This heterogeneous architecture creates challenges in designing reliable and efficient communication among the entities in the IIoT system.

Blockchain is one technology that can bring greater automation into IIoT with efficacy. There are numerous works that have employed smart contract techniques to improve operations and productivity in the IIoT sector. These blockchain-based techniques provide consensus among the various entities involved in the IIoT architecture and help in automating many human-monitored activities. The authors in [Wang et al. \(2020\)](#) have proposed a consensus mechanism that is built over the reputation of the nodes as a key attribute to determine the incentives for every valid proof

generated. [Liu et al. \(2019\)](#) proposed an efficient lightweight proof mechanism for resource-constrained nodes of IIoT systems. This mechanism reduces the workload on nodes to establish consensus among the low-power computing-enabled physical devices. They have employed a concept named “synergistic multiple proofs” that reduces the unlimited growth of a general blockchain structure to efficiently work for power-constrained networks.

Due to the ever-growing information technology-enabled critical infrastructure networks, there is always increasing concerns on the security, privacy, and reliability of these networks. [Wu et al. \(2020\)](#) have discussed the various important opportunities and challenges that are created due to the integration of blockchain and edge computing into the critical infrastructure of IIoT networks. They have proposed a layer-by-layer architecture for the blockchain- and edge-computing-enabled critical infrastructure networks. [Wan et al. \(2019\)](#) have proposed a blockchain-enabled semi-decentralized smart factory architecture to provide solutions for the privacy and security in these IIoT networks. The authors in [Guan et al. \(2020\)](#) built a decentralized two-tier architecture for IIoT networks for energy trading purposes. Their approach utilizes PoS consensus mechanisms to enable efficient and secure energy trading methods. [Kumari et al. \(2020\)](#) provided a peer-to-peer consensus mechanism through blockchain to improve the reliability, quality of service (QoS), and efficiency of the IIoT environment.

4.4 Smart cities

With the growth in the use of information technology in a multitude of domains, a smart city is no exception. Today, there is a lot of research focused to develop and deploy IT infrastructure into smart city networks. These networks involve communication among a huge number of nodes with varying degrees of characteristics and features. These nodes share various resources and require a high degree of coordination to work efficiently. Smart cities can improve the overall QoSs a user can receive. The definition of smart city networks can differ from different sources, but ultimately this architecture tries to benefit humans to improve their quality of life. Due to the complexity and high degree of nodes in a smart city, the security, reliability, and privacy of these networks remain an open challenge even today though there is significant work done ([Liu et al., 2021](#); [Mora et al., 2018](#)).

Blockchain technology has provided opportunities for research to design and develop efficient mechanisms to solve various communication- and security-related problems. [Sun et al. \(2016\)](#) proposed a blockchain-integrated, three-level framework that involves technology, humans, and organization to address the problems of reliable sharing services in the smart city scenarios. The authors in [Hakak et al. \(2020\)](#) have discussed the various characteristic features of blockchain technology and its use in smart city networks. They proposed a novel conceptual framework to provide data security that utilizes blockchain technology in the smart city network. [Sharma and Hyuk Park \(2018\)](#) proposed an innovative multidimensional framework that uses blockchain technology and software-defined networks for the smart city network. They provide the important solutions and methods that can be deployed within a smart city network to address the architectural challenges. Their consensus mechanism employs PoW concepts to establish a distributed and decentralized secure communication in a smart city network.

Data integrity and privacy is a basic requirement of smart city networks. A lot of data in these networks can be sensitive user data that requires various techniques to provide security against malicious entities. The authors in [Makhdoom et al. \(2020\)](#) present a novel privacy-preserving mechanism to protect user data from reaching attackers. They utilize a distributed blockchain network where there are different channels in these networks. Each channel contains specific entities and processes responsible for access control of the users within these networks. Their consensus mechanisms are reward-based, where they have introduced a coin named “PrivyCoin” to provide rewards for users sharing their data to different authorized entities. [Liang et al. \(2018\)](#) in their paper provide important security aspects of a smart city network. They specifically discuss in detail the scenarios of blockchain-enabled smart cities through the various attack surfaces possible. The transportation or vehicular networks are a part of the smart city. There is research work in building blockchain-enabled vehicular ad-hoc networks, to provide efficient privacy-preserving authentication, verification, and decentralization ([Lin et al., 2020](#); [Nadeem et al., 2019](#); [Wagner and McMillin, 2020](#); [Yang et al., 2019](#)).

5. Characteristics of blockchain-enabled CPS systems

[Table 3.1](#) summarizes the various characteristic properties of various blockchain-enabled CPS systems. [Lyu et al. \(2020\)](#) proposed a framework

Table 3.1 Characteristics of blockchain-enabled CPS systems.

Authors	Characteristics	Sector
Lyu et al. (2020)	• Decentralization • Access control • Data sharing • Data auditing	Information-centric networking
Gordon and Catalini (2018)	• Data aggregation • Patient identity verification • Digital access • Data immutability • Data liquidity	Healthcare industry
Xia et al. (2017)	• Data sharing • Trust-less environment • Big data • Cloud integration	Healthcare industry
Huh et al. (2017)	• Smart contract • RSA cryptography • Ethereum • Tamperproof	Internet of things
Stefan et al. (2018)	• Data integrity • Nonrepudiation • Verifiability • Decentralization	Generic CPS architecture
Xu et al. (2019)	• Authentication • Access control • Smart contract • Micro services • Smart surveillance	IoT-enhanced public safety systems
Sodhro et al. (2020)	• Convergence • Interoperability • Reliability • Decentralization	IIoT
Behzad et al. (2020)	• Sustainability • Smart contract • Energy management • Tokenization	Supply chain management
Oscar Novo (2018)	• Accessibility • Scalability • Transparency • Mobility • Concurrency • Lightweight	IoT

Continued

Table 3.1 Characteristics of blockchain-enabled CPS systems.—cont'd

Authors	Characteristics	Sector
Ibba et al. (2017)	• Data storage and management • Immutability • Availability • Smart contract	Smart cities
Lei et al. (2017)	• Distributed key management • Decentralized • Key exchange • Traffic efficiency	Intelligent transportation systems
Ismail et al. (2019)	• Lightweight • Anti-forking • Scalable • Traceable	Healthcare

for data sharing, data revocation, and audit by a user securely through the use of blockchain technology in information-centric networking. The authors in [Gordon and Catalini \(2018\)](#) have proposed mechanisms to provide patient-driven interoperability in the healthcare industry. Where this scheme can enable patient-centric data sharing without the fear of privacy concerns, it uses blockchain technology to achieve these properties. [Xia et al. \(2017\)](#) proposed a blockchain-driven control, audit, and data provenance of shared healthcare big data among the various cloud service providers. [Huh et al. \(2017\)](#) have employed blockchain-based Ethereum technology to design and develop smart contracts for IoT devices.

[Stefan et al. \(2018\)](#) have proposed a blockchain-enabled information monitoring system where the data source error in the CPS architecture can be easily detected. [Xu et al. \(2019\)](#) designed authentication and access control mechanisms using blockchain technology to provide secure decentralized data sharing micro services in IoT-enhanced public safety systems. The authors in [Sodhro et al. \(2020\)](#) have proposed a pseudo-random-based key management technique with the integration of blockchain technology for IIoT architecture to provide reliable, interoperable, and convergence of services among the various entities. [Behzad et al. \(2020\)](#) addressed the various uses of blockchain technology in SCM to provide sustainable monitoring and reporting of various activities.

[Oscar Novo \(2018\)](#) has proposed a lightweight, decentralized, distributed, blockchain-enabled system for access management in the IoT.

The authors in [Ibba et al. \(2017\)](#) have designed a “Citysense” application that utilizes blockchain technology to manage sensor data efficiently. These sensors collect environment quality indicators that include pollutants and temperature. [Lei et al. \(2017\)](#) have developed distributed key management techniques for intelligent transportation systems using blockchain technology. [Ismail et al. \(2019\)](#) proposed lightweight blockchain architecture for healthcare data management in resource-constrained networks. Their scheme does not have the forking problem as in a traditional blockchain.

6. Challenges in blockchain-enabled CPS systems

There are various challenges concerning blockchain-enabled CPSs. A few of them are listed next.

6.1 The nontechnical challenges

- There is poor knowledge transfer between academia and industry. There is a gap between the current state of theoretical research and practical implementation. There is a need to minimize the gap to get the best out of blockchain for the good of society.
- There is a lack of investment by businesses due to limited standardization of policies and practices. There are restrictions by various government entities on the use of blockchain and cryptocurrency in the public domain.
- Many existing technologies in various industries do not support migration to blockchain-enabled mechanisms. Since blockchain is evolving with time, many old legacy systems do not support newer blockchain-based technologies, making it difficult for certain organizations to adapt to the blockchain.
- The users, particularly in the healthcare sector, need to be educated before adopting blockchain technology since the data involved is sensitive in nature. Some users may agree and some may not. Largely, blockchain technology has been adopted in the public domain, and there should be caution when using blockchain for healthcare applications.
- Blockchain is not a solution to every problem. Though there is a lot of research going on regarding blockchain technology, there are many instances where blockchain is being hard fitted for some processes where it might not actually provide any positive results.

6.2 The technical challenges

- The immutability property may be a disadvantage for systems with unreliable data generation mechanisms. This property will not allow making changes to the records entered into the blockchain.
- The computing resources required for certain consensus algorithms are high. Resource constraint networks may not be able to utilize blockchain technology considering the high computational burden on the participating nodes in the network.
- Most of the research in blockchain-enabled CPS is on paper; only a few applications exist in practice.
- Some domains generate huge amounts of data, and the recording of all the data onto the blockchain can be cumbersome and costly.
- Security and privacy issues arise where blockchain records sensitive, confidential data, and user identities.
- Not every domain can adopt blockchain efficiently. There should be a detailed analysis of blockchain applications before designing systems.
- Integration and coordination of data generated by different entities of a system needs to be recorded into the blockchain in a synchronized manner.
- In some systems, there is high complexity involved while integrating blockchain technology.

6.3 Security attacks and their likelihood on blockchain-enabled CPS systems

We discuss a few security threats and their likelihood on blockchain-enabled CPS systems:

- Sniffing attack (Shi et al., 2019): The attackers in a network try the passive analysis of transaction data in blockchain to guess the identities of the user. The attack is likely to happen in a public blockchain.
- The 51% attack (Yoon, 2019): Any entity that controls 51% of computational resources or participating entities in the blockchain will control the blockchain. The attack is possible for small networks and is most unlikely for large, decentralized networks.
- The distributed denial of service (Saad et al., 2018): Attackers flood fake data to record into the blockchain network, which will exhaust resources of the network. If the block contains a hash value, then it may not impact the availability of resources to entities, but if the block contains other extra information, then each malicious entity can flood the network with fake data, likely creating an unnecessary burden on the network.

- Sybil attack (Zhang and Lee, 2019): The malicious entities create multiple identities in a network. This does not have any significant impact on the blockchain structure.
- Public ID spoofing (Hong, 2020): An attacker tries to replace the public key of other entities of a blockchain network. This attack is most unlikely to happen.
- Security vulnerabilities (de Leon et al., 2017): A poorly implemented blockchain network application is likely to suffer from various types of cyberattacks including and not limited to trojans, malware, and viruses.
- Eclipse attack (Wust and Gervais, 2016): In this attack, the attacker nodes compromise the blockchain ledger at an honest node. This is achieved when the honest node interacts with only the malicious nodes to update the blockchain in a peer-to-peer network.

7. Conclusion and future scope

In this book chapter, we first provided the preliminary background of blockchain and CPSs. Then we explained various works in blockchain-enabled CPSs in the areas of SCM, the healthcare sector, emerging industrial control systems, and smart cities. Then we stated important characteristics of blockchain-enabled CPSs considering different approaches by various authors in different areas of CPS. We discussed, in brief, the various technical and nontechnical challenges in blockchain-enabled CPSs. We have detailed some security attacks and their likelihood in the blockchain-enabled CPSs.

Blockchain is a great technology that has various important features, though the current state of practical implementation in various areas of CPS is in its infancy. There is strong scope for blockchain to evolve and be adopted at various levels in the near future. The theoretical research on blockchain uses in different areas is at its full pace.

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